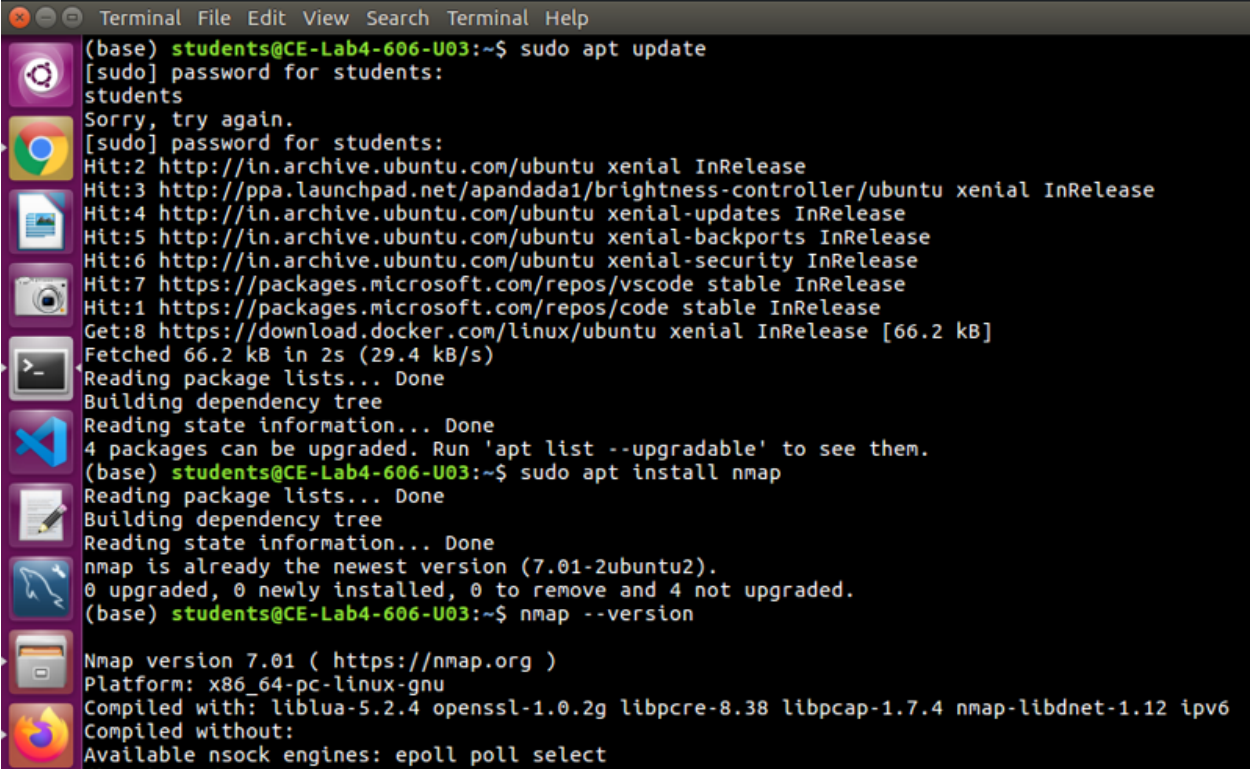


Name	Anushka Pradip Chaudhari
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Experiment No.	6

TITLE:	Network Mapping using Nmap
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PROGRAM:	Tasks: 1. Installation and Setup: ○ Install NMAP on your system if not already installed. (https://www.geeksforgeeks.org/nmap-command-in-linux-with-examples/) ○ Familiarize yourself with the basic syntax and options of NMAP. 
----------	---

2. Basic Scanning:

- Perform a simple ping scan on a target IP address to determine its availability.
- Conduct a TCP SYN scan on a target IP range to identify open ports.

```
(base) students@CE-Lab4-606-U03:~$ nmap -sn 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:36 IST
Nmap scan report for 10.10.64.25
Host is up (0.00039s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds
(base) students@CE-Lab4-606-U03:~$
```

3. Service Version Detection:

- Perform a service version detection scan on a target IP to identify the version of services running on open ports.

```
Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds
(base) students@CE-Lab4-606-U03:~$ nmap -sV 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:36 IST
Nmap scan report for 10.10.64.25
Host is up (0.00013s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE VERSION
80/tcp    open  http   Apache httpd 2.4.18 ((Ubuntu))
5900/tcp  open  vnc     VNC (protocol 3.7)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.01 seconds
(base) students@CE-Lab4-606-U03:~$
```

4. Operating System Detection:

- Use NMAP to detect the operating system of a target device.

```
Nmap done: 1 IP address (1 host up) scanned in 7.01 seconds
(base) students@CE-Lab4-606-U03:~$ sudo nmap -O 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:37 IST
Nmap scan report for 10.10.64.25
Host is up (0.00019s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE
80/tcp    open  http
5900/tcp  open  vnc
MAC Address: 24:BE:05:0E:32:64 (Hewlett Packard)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.01%E=4%D=3/6%OT=80%CT=1%CU=37718%PV=Y%DS=1%DC=D%G=Y%M=24BE05%TM
OS:=69AAA777P=x86_64-pc-linux-gnu)SEQ(SP=103%GCD=1%ISR=103%TI=Z%CI=Z%TS=A)
OS:OPS(O1=M5B4ST11NW7%O2=M5B4ST11NW7%O3=M5B4NNT11NW7%O4=M5B4ST11NW7%O5=M5B4
OS:ST11NW7%O6=M5B4ST11)WIN(W1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)
OS:ECN(R=Y%DF=Y%T=40%W=FAF0%O=M5B4NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%
OS:F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T
OS:5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=
OS:Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF
OS:=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40
OS:%CD=S)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.88 seconds
(base) students@CE-Lab4-606-U03:~$
```

5. Scripting with NMAP:

- Write a simple NMAP script to automate a scanning task of your choice.

```
nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
(base) students@CE-Lab4-606-U03:~$ nmap --script http-title 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:38 IST
Nmap scan report for 10.10.64.25
Host is up (0.000077s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE
80/tcp    open  http
|_ http-title: Apache2 Ubuntu Default Page: It works
5900/tcp  open  vnc

Nmap done: 1 IP address (1 host up) scanned in 0.22 seconds
(base) students@CE-Lab4-606-U03:~$
```

Problem Statements:

1. Scan a given network range and identify all active hosts.

```
nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
(base) students@CE-Lab4-606-U03:~$ nmap -sn 192.168.1.0/24

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:40 IST
Nmap scan report for 192.168.1.2
Host is up (0.00014s latency).
Nmap scan report for 192.168.1.3
Host is up (0.00011s latency).
Nmap scan report for 192.168.1.202
Host is up (0.00019s latency).
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.51 seconds
(base) students@CE-Lab4-606-U03:~$
```

2. Identify the top 5 most commonly open ports on a specific target.

```
(base) students@CE-Lab4-606-U03:~$ nmap --top-ports 5 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:40 IST
Nmap scan report for 10.10.64.25
Host is up (0.00029s latency).
PORT      STATE SERVICE
21/tcp    closed ftp
22/tcp    closed ssh
23/tcp    closed telnet
80/tcp    open  http
443/tcp   closed https

Nmap done: 1 IP address (1 host up) scanned in 0.02 seconds
(base) students@CE-Lab4-606-U03:~$
```

3. Determine the MAC address of a target device using NMAP.

```
(base) students@CE-Lab4-606-U03:~$ nmap -sn 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:36 IST
Nmap scan report for 10.10.64.25
Host is up (0.00039s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds
(base) students@CE-Lab4-606-U03:~$
```

4. Perform a scan to detect the presence of HTTP and HTTPS services on a target network.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 80,443 192.168.1.0/24

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:43 IST
Nmap scan report for 192.168.1.2
Host is up (0.00013s latency).
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   closed https

Nmap scan report for 192.168.1.3
Host is up (0.00011s latency).
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   closed https

Nmap scan report for 192.168.1.202
Host is up (0.00019s latency).
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Nmap done: 256 IP addresses (3 hosts up) scanned in 2.93 seconds
(base) students@CE-Lab4-606-U03:~$
```

5. Find out if a particular host has FTP service running on it.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 21 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:44 IST
Nmap scan report for 10.10.64.25
Host is up (0.00025s latency).
PORT      STATE SERVICE
21/tcp    closed ftp

Nmap done: 1 IP address (1 host up) scanned in 0.02 seconds
(base) students@CE-Lab4-606-U03:~$
```

6. Identify the SSH version running on a given host.

```
(base) students@CE-Lab4-606-U03:~$ nmap -sV -p 22 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:45 IST
Nmap scan report for 10.10.64.25
Host is up (0.00028s latency).
PORT      STATE SERVICE VERSION
22/tcp    closed ssh

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.38 seconds
(base) students@CE-Lab4-606-U03:~$
```

7. Scan a range of IP addresses and list all hosts that have Telnet service running.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 23 192.168.1.1-50

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:45 IST
Nmap scan report for 192.168.1.2
Host is up (0.00020s latency).
PORT      STATE SERVICE
23/tcp    closed telnet

Nmap scan report for 192.168.1.3
Host is up (0.00014s latency).
PORT      STATE SERVICE
23/tcp    closed telnet

Nmap done: 50 IP addresses (2 hosts up) scanned in 1.64 seconds
(base) students@CE-Lab4-606-U03:~$
```

8. Determine the operating system of a target host using NMAP.

```
(base) students@CE-Lab4-606-U03:~$ sudo nmap -O 10.10.64.25
[sudo] password for students:
students
Sorry, try again.
[sudo] password for students:

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:46 IST
Nmap scan report for 10.10.64.25
Host is up (0.00020s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE
80/tcp    open  http
5900/tcp  open  vnc
MAC Address: 24:BE:05:0E:32:64 (Hewlett Packard)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.01%E=4%D=3/6%OT=80%CT=1%CU=39624%PV=Y%DS=1%DC=D%G=Y%M=24BE05%TM
OS:=69AA996P=x86_64-pc-linux-gnu)SEQ(SP=103%GCD=1%ISR=108%TI=Z%CI=Z%TS=A)
OS:OPS(O1=M5B4ST11NW7%O2=M5B4ST11NW7%O3=M5B4NNT11NW7%O4=M5B4ST11NW7%O5=M5B4
OS:ST11NW7%O6=M5B4ST11)WIN(W1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)
OS:ECN(R=Y%DF=Y%T=40%W=FAF0%O=M5B4NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%
OS:F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=RD=0%Q=)T
OS:5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+F=AR%O=RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=
OS:Z%F=R%O=RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=Z%A=S+F=AR%O=RD=0%Q=)U1(R=Y%DF
OS:=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40
OS:%CD=S)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.79 seconds
(base) students@CE-Lab4-606-U03:~$
(base) students@CE-Lab4-606-U03:~$
```

9. Identify any SQL services running on a given network.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 3306,1433,5432 192.168.1.0/24

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:47 IST
Nmap scan report for 192.168.1.2
Host is up (0.00011s latency).
PORT      STATE SERVICE
1433/tcp   closed ms-sql-s
3306/tcp   closed mysql
5432/tcp   closed postgresql

Nmap scan report for 192.168.1.3
Host is up (0.000098s latency).
PORT      STATE SERVICE
1433/tcp   closed ms-sql-s
3306/tcp   closed mysql
5432/tcp   closed postgresql

Nmap scan report for 192.168.1.202
Host is up (0.00019s latency).
PORT      STATE SERVICE
1433/tcp   filtered ms-sql-s
3306/tcp   filtered mysql
5432/tcp   filtered postgresql

Nmap done: 256 IP addresses (3 hosts up) scanned in 3.73 seconds
(base) students@CE-Lab4-606-U03:~$
```

10. Find out if a specific host has Remote Desktop Protocol (RDP) enabled.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 3389 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:48 IST
Nmap scan report for 10.10.64.25
Host is up (0.00028s latency).
PORT      STATE SERVICE
3389/tcp   closed ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
(base) students@CE-Lab4-606-U03:~$
```

11. Scan a target network and determine if any hosts are running DNS services.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 53 192.168.1.0/24

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:48 IST
Nmap scan report for 192.168.1.2
Host is up (0.00026s latency).
PORT      STATE SERVICE
53/tcp     closed domain

Nmap scan report for 192.168.1.3
Host is up (0.00021s latency).
PORT      STATE SERVICE
53/tcp     closed domain

Nmap scan report for 192.168.1.202
Host is up (0.00026s latency).
PORT      STATE SERVICE
53/tcp     filtered domain

Nmap done: 256 IP addresses (3 hosts up) scanned in 2.96 seconds
(base) students@CE-Lab4-606-U03:~$
```

12. Detect if a host has SNMP (Simple Network Management Protocol) enabled.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 161 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:48 IST
Nmap scan report for 10.10.64.25
Host is up (0.00027s latency).
PORT      STATE SERVICE
161/tcp    closed snmp

Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds
(base) students@CE-Lab4-606-U03:~$
```

13. Perform a scan to identify any SMTP (Simple Mail Transfer Protocol) servers on a network.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 25 192.168.1.0/24

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:49 IST
Nmap scan report for 192.168.1.2
Host is up (0.00013s latency).
PORT      STATE SERVICE
25/tcp    closed smtp

Nmap scan report for 192.168.1.3
Host is up (0.00011s latency).
PORT      STATE SERVICE
25/tcp    closed smtp

Nmap scan report for 192.168.1.202
Host is up (0.00019s latency).
PORT      STATE SERVICE
25/tcp    filtered smtp

Nmap done: 256 IP addresses (3 hosts up) scanned in 2.63 seconds
(base) students@CE-Lab4-606-U03:~$
```

14. Determine if a target network has any active FTP servers allowing anonymous login.

```
(base) students@CE-Lab4-606-U03:~$ nmap --script ftp-anon -p 21 192.168.1.0/24

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:49 IST
Nmap scan report for 192.168.1.2
Host is up (0.00013s latency).
PORT      STATE SERVICE
21/tcp    open  ftp

Nmap scan report for 192.168.1.3
Host is up (0.00011s latency).
PORT      STATE SERVICE
21/tcp    open  ftp

Nmap scan report for 192.168.1.202
Host is up (0.00026s latency).
PORT      STATE SERVICE
21/tcp    filtered ftp

Nmap done: 256 IP addresses (3 hosts up) scanned in 3.30 seconds
(base) students@CE-Lab4-606-U03:~$
```

15. Find out if any hosts in a network are running vulnerable versions of the Apache HTTP server.

```
(base) students@CE-Lab4-606-U03:~$ nmap -sV --script http-vuln* -p 80 192.168.1.0/24

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:50 IST
Nmap scan report for 192.168.1.2
Host is up (0.00013s latency).
PORT      STATE SERVICE
80/tcp    open  http   Apache httpd
|_http-server-header: Apache

Nmap scan report for 192.168.1.3
Host is up (0.00010s latency).
PORT      STATE SERVICE
80/tcp    open  http   Apache httpd
|_http-server-header: Apache

Nmap scan report for 192.168.1.202
Host is up (0.00025s latency).
PORT      STATE SERVICE
80/tcp    open  http   Apache httpd 2.4.18 ((Ubuntu))
|_http-server-header: Apache/2.4.18 (Ubuntu)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (3 hosts up) scanned in 9.93 seconds
(base) students@CE-Lab4-606-U03:~$
```

16. Detect if a target host has any open NFS (Network File System) shares.

```
Nmap done: 256 IP addresses (3 hosts up) scanned in 9.93 seconds
(base) students@CE-Lab4-606-U03:~$ nmap -p 2049 --script nfs-showmount 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:51 IST
Nmap scan report for 10.10.64.25
Host is up (0.00034s latency).
PORT      STATE SERVICE
2049/tcp  closed nfs

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
(base) students@CE-Lab4-606-U03:~$
```

17. Identify the presence of any MySQL database servers on a given network.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 3306 192.168.1.0/24

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:51 IST
Nmap scan report for 192.168.1.2
Host is up (0.00013s latency).
PORT      STATE SERVICE
3306/tcp  closed mysql

Nmap scan report for 192.168.1.3
Host is up (0.00010s latency).
PORT      STATE SERVICE
3306/tcp  closed mysql

Nmap scan report for 192.168.1.202
Host is up (0.00030s latency).
PORT      STATE SERVICE
3306/tcp  filtered mysql

Nmap done: 256 IP addresses (3 hosts up) scanned in 2.83 seconds
(base) students@CE-Lab4-606-U03:~$
```

18. Scan a network to determine if any hosts have the Remote Procedure Call (RPC) service running.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 111 192.168.1.0/24

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:52 IST
Nmap scan report for 192.168.1.2
Host is up (0.00023s latency).
PORT      STATE SERVICE
111/tcp    closed rpcbind

Nmap scan report for 192.168.1.3
Host is up (0.00018s latency).
PORT      STATE SERVICE
111/tcp    closed rpcbind

Nmap scan report for 192.168.1.202
Host is up (0.00029s latency).
PORT      STATE SERVICE
111/tcp    filtered rpcbind

Nmap done: 256 IP addresses (3 hosts up) scanned in 2.53 seconds
(base) students@CE-Lab4-606-U03:~$
```

19. Detect if a specific host has any open VNC (Virtual Network Computing) ports.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p 5900,5901 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:52 IST
Nmap scan report for 10.10.64.25
Host is up (0.00035s latency).
PORT      STATE SERVICE
5900/tcp   open  vnc
5901/tcp   closed vnc-1

Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
(base) students@CE-Lab4-606-U03:~$
```

20. Perform a scan to identify any hosts with the Secure Shell (SSH) service running on non-default ports.

```
(base) students@CE-Lab4-606-U03:~$ nmap -p- --open 10.10.64.25 | grep ssh
(base) students@CE-Lab4-606-U03:~$ nmap -p- --open 192.168.1.10

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:54 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.05 seconds
(base) students@CE-Lab4-606-U03:~$ nmap -p- --open 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:54 IST
Nmap scan report for 10.10.64.25
Host is up (0.00011s latency).
Not shown: 65533 closed ports
PORT      STATE SERVICE
80/tcp    open  http
5900/tcp  open  vnc

Nmap done: 1 IP address (1 host up) scanned in 1.27 seconds

(base) students@CE-Lab4-606-U03:~$ nmap -sV -p- 10.10.64.25

Starting Nmap 7.01 ( https://nmap.org ) at 2026-03-06 15:55 IST
Nmap scan report for 10.10.64.25
Host is up (0.000090s latency).
Not shown: 65533 closed ports
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.18 ((Ubuntu))
5900/tcp  open  vnc       VNC (protocol 3.7)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.49 seconds
(base) students@CE-Lab4-606-U03:~$
```

CONCLUSION:

The Nmap scan was successfully performed to detect SSH services running on non-default ports. The results showed that no SSH service was active on any port other than the default port 22. Only HTTP and VNC services were identified on the target system. Hence, it can be concluded that the target host does not have SSH configured on non-standard ports.